

GAURAB BISHWAKARMA

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🌐 [MyPortfolio](#) | [GitHub](#) | [LinkedIn](#)

Objective

Final-year BCA student with practical experience in full-stack development (MERN, WebRTC, Socket.io). I built AksharShelf, a web reading platform, and worked on Metra, a video conferencing app, handling everything from backend logic to the real-time media pipeline. Looking for an internship to grow as a developer and contribute to real projects.

Technical Skills

Frontend: HTML, CSS, JavaScript, React, Tailwind CSS

Backend: Node.js, Express.js, MongoDB, Socket.io, WebRTC, Mediasoup, RESTful APIs

Tools & Technologies: Git, GitHub, Postman, VS Code

Projects

AksharShelf – Web Reading Platform

[Code](#) | [Live Demo](#)

Tech Stack: React, Node.js, Express.js, MongoDB, Tailwind CSS, JWT

- **Full-Stack Development:** Built a comprehensive web reading platform using the MERN stack featuring book discovery, bookshelf management, and user reviews.
- **User Experience:** Implemented a personalized recommendation system and user profiles with JWT-based authentication for seamless book browsing.
- **Admin Dashboard:** Developed an administrative panel for content management, user management, and platform analytics.
- **Architecture:** Applied structured software development principles with detailed system architecture diagrams, DFD, and algorithm documentation.

Metra – Video Conference Web App

[Code](#) (Team Project – In Progress)

Tech Stack (My Part): WebRTC, Mediasoup, Socket.io

- **Real-Time Communication:** Engineered the core WebRTC and Mediasoup-based video and audio streaming pipeline, enabling low-latency peer-to-peer and SFU media routing for multiple concurrent participants.
- **Signaling & Presence:** Implemented Socket.io signaling server to handle room creation, participant join/leave events, and media negotiation between client and media server.
- **Media Server Integration:** Configured Mediasoup workers, routers, and transports to support scalable multi-party video conferencing with dynamic producer and consumer management.

Education

Bachelor of Computer Application (BCA)

Patan Multiple Campus, Tribhuvan University

2022 – Present

Patandhoka, Lalitpur